1. Clinical Question & Core Hypothesis

Clinical Question

NMIBC shows heterogeneous non-invasive biomarker signals and variable responses to intravesical BCG; why do some tumors with apparent local immune organization (TLS/TILs) still fail to respond or shed detectable tumor material into blood/stool? The Ontologist suggests discordance arises from interactions among tumor shedding (ADNP in CTCs), tumor mutational/repair state (MSH6, bTMB), and local immune architecture. 2) Tier■1 paths used: I built on CORE PATH 1 (ADNP mRNA detectable in CTCs → liquid biopsy signal) because it gives a concrete, measurable tumor■shedding payload; CORE PATH 3 (MSH6 as prognostic/MMR factor) because mismatch■repair dysfunction mechanistically alters mutation load and antigen quality; and CORE PATH 2 (BCG → PD■L1/bTMB linkage) as mechanistic context linking local therapy to immune/genomic readouts. These together create a testable mechanistic puzzle: when and why does local immune structure (TLS) fail to translate into therapy response or reliable liquid■biopsy detection? 3) Tier■2 evidence support: GARS1 suggests tumor metabolic/translation states may modulate immune surveillance and shedding; SDC2/TFPI2 methylation highlights complementary non■invasive epigenetic readouts that could validate shedding versus local release; systemic metabolic markers and AI■based accessibility inform modifier variables and implementation feasibility. 4) Endpoint choice: therapy_response (BCG response) was chosen because BCG is the frontline, immune■mediated treatment in NMIBC, and the ontologist's core paths explicitly link BCG/PD■L1/bTMB dynamics to biomarker performance; predicting BCG response is clinically actionable and mechanistically tied to the immune/genomic variables under study. 5) Exclusions/deliberate simplifications: I excluded broader exploratory paths (Immunoscore, SARIFA, ER/TP53 associations, systemic metabolic panels) from the core hypothesis to keep focus on a single imaging feature + two anchor genes; those appear in secondary_modifiers or are left for follow■up analyses because they are either cross■cancer translations or multi■factorial and would dilute mechanistic clarity?

Working Hypothesis

TLS-like lymphoid aggregates are not universally predictive of intravesical BCG response in NMIBC. Their predictive value is contingent on two tumor-derived states: (1) ADNP-enriched circulating tumor cell (CTC) shedding, which reflects systemic tumor payload and may indicate dynamic antigen release/immune interaction, and (2) MSH6 status-driven mismatch repair dysfunction, which elevates mutational load and neoantigen quality. We hypothesize that: a) in patients with low ADNP-CTC shedding and intact MSH6, TLS maturity correlates positively with BCG response; b) in patients with high ADNP-CTC shedding and/or MSH6 deficiency (high bTMB), TLS presence fails to predict response due to altered antigen presentation and potential immune escape; c) a composite biomarker integrating TLS metrics (density and maturation), ADNP-CTC signal, and MSH6/bTMB will predict BCG responders vs non-responders with higher accuracy than any single marker. This model implies mechanistic interactions where local immune organization is necessary but not sufficient, and systemic tumor payload plus repair-defective mutational landscapes modulate whether TLS translates into effective anti-tumor immunity. Predictions include: (i) TLS maturity alone will predict response only in MSH6-intact and low ADNP-CTC contexts; (ii) ADNP-CTC positivity will associate with non-response even when TLS is present; (iii) the best discriminatory power will come from a combined scoring system (TLS maturity × ADNP-CTC status × MSH6/bTMB) with superior AUC over single-parameter models; (iv) non-invasive methylation markers (SDC2, TFPI2) will parallel shedding signals and reinforce the interpretation of ADNP-CTC data. Feasibility-wise, this requires concurrent tissue TLS assessment, CTC RNA analysis, germline/sporadic MSH6 evaluation, bTMB calculation, and cfDNA methylation profiling in a prospective NMIBC cohort undergoing BCG.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	ADNP, MSH6
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ADNP: ADNP mRNA detectable in CTCs → liquid biopsy signal) because it gives a concrete, measurable tumor■shedding payload; CORE PATH 3 (MSH6 as prognostic/MMR factor) because mismatch■repair dysfunction mechanistically alters mutation load and antigen quality; and

CORE PATH 2 (BCG → PD■L1/bTMB linkage) as mechanistic context linking local therapy to immune/genomic readouts.

MSH6: MSH6 as prognostic/MMR factor) because mismatch■repair dysfunction mechanistically alters mutation load and antigen quality; and CORE PATH 2 (BCG → PD■L1/bTMB linkage) as mechanistic context linking local therapy to immune/genomic readouts.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PD-L1 (CD274)	Supporting modifier from knowledge graph
Secondary	bTMB	Supporting modifier from knowledge graph
Secondary	GARS1	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: offer complementary evidence of shedding and tumor-derived epigenetic alterations. Integrating these layers addresses the mechanistic gap between local immune architecture and systemic immune–tumor dynamics, with potential to refine patient stratification for BCG or alternative therapies.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PD-L1 (CD274): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- bTMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.